One-pot PCR (head-tailed gene-region primers + barcoded-head primers, one tube)

HuangLab (Yin-Tse Huang lab, NCKU)

(v1.0, 2026-10-20; 50 µL reaction, KAPA HiFi HotStart)

Source: lab sheet *ONE POT PCR.xlsx*. Product = *barcode – head – gene region – head – barcode*, same as the two-step route. One tube per sample, one barcode per tube. Set for ~3–4 kb amplicons.

Use when: many samples, same region, templates that already amplify well. **Not for** inhibitor-rich or low-copy templates: there is no 1st-PCR product to check.

Materials :

1. 2X KAPA HiFi HotStart ReadyMix
2. Head-tailed gene-region primers F + R, 10 µM (same as two-step 1st PCR)
3. Pre-mixed barcoded-head primer pairs, 10 µM each (barcode plate)
4. DEPC-treated water
5. DNA template, 1–10 ng/µL, MolWork# assigned
6. 8-strip PCR tubes / 96-well plate, filter tips, 1% bleach

※ **Note:** Barcoded-head primers must carry the same head sequence as the gene-region primer tails.

Reaction mixture (per 50 µL)

Master mix = ReadyMix + gene-region primers + water, 1.2X for $n + 1$ reactions (negative control included). Add barcode pair and template per tube.

Component	Volume	Final
2X KAPA HiFi HotStart ReadyMix	**25 µL**	1X
Gene-region primer F, head-tailed (10 µM)	**0.5 µL**	**0.1 µM**
Gene-region primer R, head-tailed (10 µM)	**0.5 µL**	**0.1 µM**
DEPC-treated water	**19.5 µL**	
Master mix subtotal	***45.5 µL***	
Barcoded-head primers F+R (10 µM each)	**1.5 µL**	**0.3 µM each**
DNA template	**3 µL**	
Total	**50 µL**	

Procedure

Setup

1. Clean bench with 1% bleach. Thaw reagents on ice, vortex, spin down.

2. Make master mix. Mix by pipetting, spin down.

3. Dispense **45.5 µL** master mix per tube.

4. Add **1.5 µL** barcoded-head primer pair, **one barcode per tube**. Write the barcode on the tube and in

the LIMS row.

5. Add **3 µL** template. Negative control: **3 µL** water.

6. Tap to mix, spin down, keep on ice.

Cycling program

7. Phase 1 anneals at the gene-region primer Ta; phase 2 at **65 °C** (barcoded-head primers).

Stage	Step	Temp	Time	Cycles
1	Initial denaturation	**98 °C**	**45 s**	**1×**
2	Denaturation	**98 °C**	**10 s**	**25×**
2	Annealing	**60–66 °C** (a)	**30 s**	
2	Extension	**72 °C**	**2 min 30 s (b)**	
3	Intermediate extension	**72 °C**	**5 min**	**1×**
4	Denaturation	**98 °C**	**30 s**	**1×**
5	Denaturation	**98 °C**	**15 s**	**15×**
5	Annealing	**65 °C**	**15 s**	
5	Extension	**72 °C**	**30 s**	
6	Final extension	**72 °C**	**2 min 30 s**	**1×**
7	Hold	**4 °C**	∞	

(a) From the primer list; LIMS method = one_pot_60 / 62 / 64. (b) ~1 min per kb.

※ **Note:** 40 cycles total. Do not add phase-2 cycles (chimeras, primer dimers).

QC and registration

8. Gel: **2 µL** per lane, 1% agarose. One band at expected size; negative control blank. Head-only extra band → phase 2 incomplete. Band in negative control → discard batch.

9. Photograph gel, mark wells, upload to lab Google Drive, record pcr_ok.

10. Qubit Flex (dsDNA HS) each tube; enter concentration in LIMS.

11. LIMS Stage 3b (/lims-register): one pcr_reaction row per MolWork#, method = one_pot_60 / 62 / 64, taq = KAPA HiFi HotStart. Assign flow cell + barcode at setup; the single OK makes the sample ready for Stage 4a.

12. Pool equal molar → *Size selection (1X)* clean-up → EvoPrep library.